
Tutorial 2

Foundations of Cryptography

Computer Security

BCS2420

Part A

1. Which of the following best describes the advantage of **public-key** (asymmetric) encryption compared to **symmetric encryption**?
 - ☐ Faster encryption for large data
 - ☐ The same key is used by both sender and receiver
 - ☐ It simplifies secure key distribution
 - ☐ It eliminates the need for any private key
2. Which cryptographic property ensures that a given hash output does not reveal any information about the original message?
 - ☐ Collision resistance
 - ☐ One-way property
 - ☐ Key encapsulation
 - ☐ Second-preimage resistance
3. In a chosen-ciphertext attack model, the adversary can...
 - ☐ Only observe ciphertexts without modifying them
 - ☐ Submit ciphertexts to a decryption oracle and observe the results
 - ☐ Pick plaintext and obtain its encryption from an oracle
 - ☐ Obtain a set of plaintext-ciphertext pairs without influencing them
4. Which statement about the *Vernam cipher* is **correct**?
 - ☐ It encrypts data in 64-bit blocks.
 - ☐ It provides perfect secrecy if the key is reused carefully.

- ☐ It uses modular exponentiation with prime numbers.
 - ☐ It can provide perfect secrecy only if the key is truly random and used exactly once.
5. **In a hybrid encryption scheme, typically...**
- ☐ Only the public key is needed to encrypt large messages directly.
 - ☐ Symmetric keys are never involved.
 - ☐ A symmetric key is randomly generated for data encryption, then encrypted with a public key.
 - ☐ The private key is used for both encryption and decryption.
6. **Which of the following is an *active* adversary action?**
- ☐ Eavesdropping on network traffic without modifying it
 - ☐ Recording ciphertext for later analysis
 - ☐ Injecting malicious packets into a conversation
 - ☐ Archiving all messages for offline cryptanalysis
7. **Collision resistance in a hash function means...**
- ☐ Finding two distinct inputs with the same hash is computationally infeasible.
 - ☐ Reconstructing the original message from the hash is infeasible.
 - ☐ Finding any input that produces a given hash is easy.
 - ☐ Only one hash value is possible for a given input.
8. **Which block cipher mode directly concatenates each plaintext block with the preceding ciphertext block (via XOR) before encryption?**
- ☐ ECB
 - ☐ CBC
 - ☐ CTR
 - ☐ OFB
9. **Digital signatures aim to provide which of the following?**
- ☐ Confidentiality and authentication only
 - ☐ Non-repudiation, integrity, and origin authentication
 - ☐ Key management and collision resistance
 - ☐ Simple encryption of large files
10. **A 56-bit key space (as in original DES) implies how many possible keys?**
- ☐ 2^{56}
 - ☐ 56^2
 - ☐ 10^{56}
 - ☐ 2×10^{16}

Part B

1. **Comparing Passive vs. Active Adversaries:**
 - Briefly define **passive** and **active** adversaries in cryptography.
 - Provide one real-world example scenario of each.
 - Why might defending against active adversaries be more challenging?
2. **Attack Models:**
 - Summarize the four attack models discussed (ciphertext-only, known-plaintext, chosen-plaintext, chosen-ciphertext).
 - Which model imposes the *strongest* requirement on a cryptosystem's security, and why?
3. **Key Space and Security:**
 - Explain the concept of **key space**.
 - Discuss how an increase in key length affects the feasibility of **brute-force** attacks.
 - Give two *practical factors* (beyond key length) that also determine a cipher's real-world resistance to brute-force.
4. **Designing a Secure Symmetric Cipher:**
 - Suppose you aim to design a new symmetric encryption system. What *core properties* must it have to resist modern cryptanalytic attacks?
 - Mention any **two** advanced cryptanalytic methods and how a well-designed cipher can mitigate them.
5. **Public-Key Infrastructure (PKI):**
 - What is a **certificate** and how does it relate to **trust** in PKI?
 - Illustrate how an organization could use PKI to ensure secure email communication among its employees.
6. **Hash Functions vs. Encryption Functions:**
 - Compare the roles of a *cryptographic hash function* and a *symmetric encryption function*.
 - Why is it generally inappropriate to use a **block cipher** in place of a hash function?
7. **Digital Signatures and Non-Repudiation:**
 - What does "**non-repudiation**" mean in the context of

digital signatures?

- Give an example in an e-commerce transaction where non-repudiation is crucial.

8. **Block Cipher Modes:**

- Discuss the main weakness of **ECB** mode.
- Why is **CBC** mode generally preferred over ECB for encrypting multiple blocks?
- Provide a high-level use-case for **CTR** (counter) mode.

9. **Hybrid Encryption Scenario:**

- You want to encrypt a 1GB file for secure storage. Why might you opt to use a **hybrid** encryption approach instead of directly using a public-key cipher for the entire file?
- Summarize the key steps in hybrid encryption for this scenario.

10. **MAC vs. Digital Signature:**

- In which scenarios is a **Message Authentication Code (MAC)** more suitable than a **digital signature**?
 - Conversely, give an example scenario where a digital signature is more beneficial.
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Part C

1. **Brute-Force Feasibility**

- A cryptographic system has a key space of 2^{40} .
- How many keys are there in total?
- If an attacker can try 10^9 (1 billion) keys per second, how many **seconds** to exhaust the key space in the worst case? Convert this to **days** (approx.).
- Comment on whether this is feasible with today's technology.

2. **Key Length Impact**

- You are deciding between a 128-bit key and a 256-bit key.
- By how many *orders of magnitude* does a 256-bit key space exceed a 128-bit key space?
- Does doubling the key length from 128 bits to 256 bits *double* the security, or provide more? Briefly explain.

3. **Simple Caesar Cipher Computation**

- The Caesar cipher shifts letters by **3**. Given the ciphertext "FODCGR" (assuming A=0, B=1, ... Z=25), decrypt it back to plaintext.
- Show your steps.
- How many possible shifts (keys) are there in total if we only consider the English alphabet?

4. **Block Cipher Encryption Rate**

- A block cipher encrypts **128-bit** blocks. The implementation can process **1 million** blocks per second.
- How many **bits per second** does this correspond to?
- How many **megabytes per second** is this (assuming 1 byte = 8 bits)?
- How long would it take to encrypt a **10 GB** file?

5. **Hash Collisions**

- A cryptographic hash has an **n-bit** output. By the birthday paradox, collisions typically appear after about $\sqrt{2^n}$ operations.
- For a **128-bit** hash, how many operations (roughly) might be needed to find a collision?
- Why might attackers find collisions sooner if the hash function is poorly designed?

6. **Public-Key Encryption Overhead**

- An organization wants to transmit **10,000** messages, each 1KB in size, using pure **public-key** encryption with RSA (no hybrid approach). Each RSA encryption takes **5ms**.
 - How long (in seconds) will it take to encrypt all messages?
 - Why might the organization prefer hybrid encryption in this scenario?
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